

Supplemental Material for:
**“Photon-efficient neural inference over optical links: training and
adaptive operation at the quantum limit”**

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Abstract

Overview and provenance. This document provides the complete methods and the complete numerical record behind the Letter: the model, data, training, and controller details; the twin-validation table against the analytic standard quantum limit (SQL); the photon-budget accounting; the full accuracy sweeps for every dataset, training protocol, chain condition, and photon budget (with standard deviations over the five evaluation seeds); the individual impairment sweeps; the quantitative comparison with the state of the art; the complete photon-budget Pareto sets of the closed-loop controller together with the oracle escalation bounds; and the training hyperparameters. Every entry is generated programmatically from the released results record (`results.json` in the Zenodo archive of the Letter’s Data Availability statement); no value is typed by hand. Bracketed reference numbers refer to the Letter’s reference list.

S-I. METHODS

A. Tasks, models, and data

Three open benchmarks are used: MNIST (60,000 train/10,000 test), the common benchmark of the photon-starved literature [1–3]; Fashion-MNIST (same split; strictly harder); and the Free Spoken Digit Dataset (FSDD) of 3,000 recordings at 8 kHz, processed by an amplitude short-time-Fourier front-end into 4,000-dimensional nonnegative vectors (speaker-stratified split, 2,700/300). Models are real-valued fully connected networks matched to the thin-client architecture: intensity-encoded nonnegative inputs, signed weights on differential rails, and ReLU between layers (784–300–100–10 for images; 4000–300–100–10 for audio).

B. Training protocols

All models train with Adam (10^{-3} , decayed $\times 0.3$ at 70% of training, batch 256) in PyTorch. Three protocols differ only in the forward physics: *conventional* (clean digital arithmetic, the practice of the delocalized demonstration [1]); *PILOT-Q* (each batch tra-

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verses the twin at $\bar{n} \sim \log \mathcal{U}[0.25, 8]$ photons/MAC; $\log \mathcal{U}[0.5, 8]$ on Fashion-MNIST, whose harder decision boundary tolerates less training noise); and *PILOT-Q-full* (additionally $\rho_d \sim \mathcal{U}[0, 0.20]$, $\sigma_{cal} \sim \mathcal{U}[0, 0.15]$, and $b \in \{4, 6, 8, \infty\}$ per batch). The physics-in-the-loop protocols anneal into the physics (first 15% of epochs clean; 35% of batches keep the clean anchor). Image models train 25/35 epochs (conventional/physics-in-the-loop); audio models 100/140. Evaluation draws fresh noise with five fixed seeds; $\pm$ values are standard deviations across seeds. A fixed-budget ablation (training at a constant $\bar{n} = 1$) is reported in Fig. S2(d) and Table SV.

C. Closed-loop controller

At run time the client first measures at a cheap budget $\bar{n}_{lo}$ and computes the softmax margin $m = p_{(1)} - p_{(2)}$. If $m \geq \tau$ the prediction commits; otherwise the client re-exposes at $\bar{n}_{hi}$ and pools the two estimates with photon-number weights, $\hat{\mathbf{y}} = (\bar{n}_{lo}\hat{\mathbf{y}}_1 + \bar{n}_{hi}\hat{\mathbf{y}}_2)/(\bar{n}_{lo} + \bar{n}_{hi})$, which is the maximum-likelihood combination of two independent exposures, so no photon is wasted. The mean budget is $\bar{\Phi} = \Phi_{lo} + p_{esc}\Phi_{hi}$; sweeping τ traces an accuracy–photon Pareto front. To delimit what any policy of this form can achieve, an oracle bound is also computed: escalate exactly the inputs mispredicted at $\bar{n}_{lo}$, an upper envelope that no realizable confidence signal can exceed.

S-II. TWIN VALIDATION AT THE STANDARD QUANTUM LIMIT

Figure S1(a) and Table SIII report the Monte Carlo normalized inner-product RMSE of the twin at every evaluated photon budget for both problem sizes, next to the analytic SQL prediction of the differential-rail detection model [Eq. (1) of the Letter]. Unlike an empirically fitted noise model, agreement here is a check against theory: the twin matches the closed-form shot-noise law within Monte Carlo error across three decades of $\bar{n}$, with exact $1/\sqrt{\bar{n}}$ scaling.

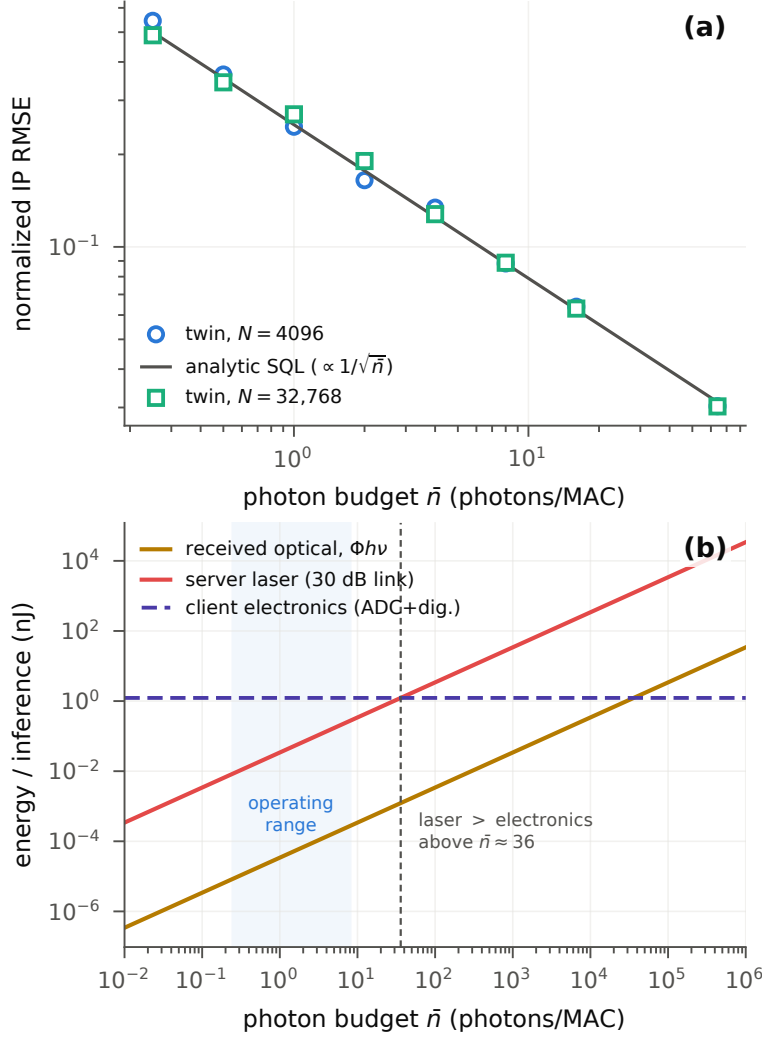


FIG. S1. Twin validation and photon-budget accounting. (a) Normalized inner-product RMSE of the twin versus photon budget at $N=4096$ and $N=32,768$, against the analytic standard-quantum-limit law (solid): agreement within 9% over three decades, with exact $1/\sqrt{\bar{n}}$ scaling. (b) Per-inference energy accounting for the 784-input model: received optical energy $\Phi h\nu$ and the server laser energy through a 30 dB link budget scale linearly with $\bar{n}$; client electronics (ADC + decode at the reference constants of the wireless in-physics study [4]) are photon-independent and dominate the client at every practical budget. The shaded band marks the operating range studied.

S-III. PHOTON-BUDGET ACCOUNTING

Table [SIV](#) tabulates the accounting shown in Fig. [S1\(b\)](#) and summarized in the Letter: the received photon budget $\Phi = \bar{n} \times \text{MACs}$ (266,200 MACs for the image models; 1,231,000 for audio) and its optical energy, the server laser energy through a representative 30 dB link (scale linearly for other link budgets), and the photon-independent client electronics at the reference constants (1 pJ per ADC sample and per digital op) adopted by the wireless in-physics literature [4]. The client is electronics-limited at every budget studied; the photon budget prices the server laser, the aggregate broadcast, and the integration time.

S-IV. FULL ACCURACY SWEEPS AND SUMMARY TABLES

Tables [SV](#)–[SVII](#) report the complete accuracy versus photon-budget data underlying Fig. 2 of the Letter: every dataset, protocol (including the fixed-budget ablation on MNIST), photon budget, and both chain conditions, as mean $\pm$ standard deviation over the five fixed seeds. The final row of each table gives the clean digital ceiling of each protocol. Table [SI](#) consolidates the accuracy summary, and Table [SII](#) the iso-accuracy photon-budget accounting of the closed loop.

S-V. INDIVIDUAL IMPAIRMENT SWEEPS

Figure [S2](#) dissects the impairments at $\bar{n} = 2$ photons/MAC on MNIST, and Tables [SVIII](#)–[SX](#) report the dark-count, trim-error, and ADC-resolution sweeps for all datasets and protocols. As in analog in-memory computing [14, 15], the per-element and nonlinear impairments are where retraining restructures the solution; against pure shot noise, which is signal-proportional by physics, the conventional solution is already near-optimal at generous budgets, and the PILOT-Q advantage concentrates in the photon-starved tail, consistent with the bench-top photon-starved literature [2, 3]. The ablation in Fig. [S2\(d\)](#) shows the robustification is not an artifact of distributional training: fixed- $\bar{n}$ training captures most of it, and the randomized band generalizes better across budgets.

TABLE SI. Master accuracy summary. Ideal: clean digital chain. Shot-limited and impaired ($\rho_d=20\%$, $\sigma_{cal}=15\%$, 4-bit ADC) accuracies at two photon budgets. $\bar{n}_{tgt}$: budget at which the shot-limited accuracy reaches the dataset target (two points below the conventional ceiling); Φ_{tgt} : corresponding per-inference photon budget.

Dataset	Training	Ideal	Shot-limited acc. (%)		Impaired acc. (%)		$\bar{n}_{tgt}$	Φ_{tgt}
		acc. (%)	0.5 ph	2 ph	0.5 ph	2 ph	(ph/MAC)	(Mphotons)
MNIST	conventional	98.3	95.8	97.8	89.8	95.8	0.63	0.17
	PILOT-Q	98.2	96.8	98.0	94.7	97.2	0.45	0.12
	PILOT-Q-full	97.1	92.8	96.2	89.2	94.9	2.31	0.61
Fashion-MNIST	conventional	89.6	75.6	87.0	62.9	79.2	2.72	0.72
	PILOT-Q	89.3	83.1	87.9	78.0	85.9	1.73	0.46
	PILOT-Q-full	86.3	76.2	83.7	71.5	81.5	–	–
FSDD	conventional	89.7	47.0	75.1	39.3	61.3	13.38	16.47
	PILOT-Q	88.0	75.3	86.6	68.7	83.3	4.29	5.28
	PILOT-Q-full	85.7	53.9	74.5	49.3	67.9	–	–

TABLE SII. Iso-accuracy photon-budget accounting per inference. Static columns: operation at the budget where each protocol reaches the dataset target on the shot-limited chain; closed loop: the photon-cheapest $(\bar{n}_{lo}, \bar{n}_{hi}, \tau)$ point reaching the target with the PILOT-Q model. Server laser energy at a 30 dB link budget; client electronics are photon-independent.

Dataset	Photon budget at target (Mphotons/inf.)			red.	Server laser	Eff. $\bar{n}$	Esc.
	conv.	PILOT-Q	closed loop	(conv. base)	(pJ @ 30 dB)	(ph/MAC)	(%)
MNIST	0.17	0.12	0.13	$1.3\times$	17.1	0.50	0.0
Fashion-MNIST	0.72	0.46	0.53	$1.4\times$	68.2	2.00	0.0
FSDD	16.47	5.28	4.98	$3.3\times$	637.8	4.04	0.3

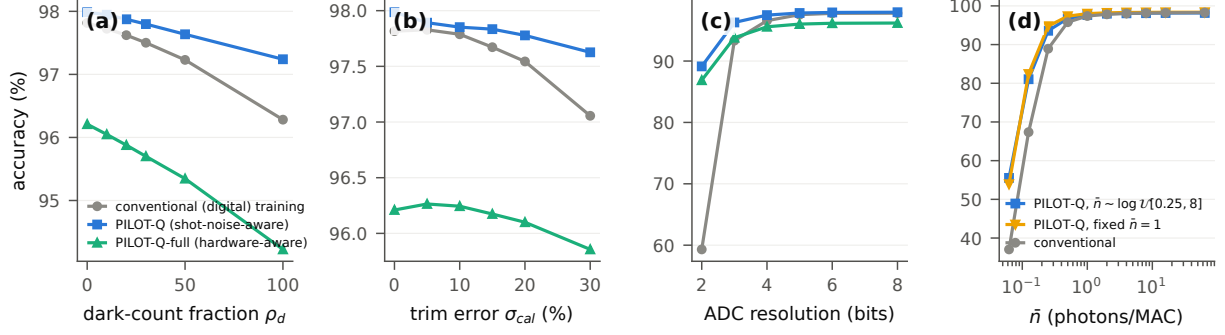


FIG. S2. Impairment dissection (MNIST, $\bar{n}=2$ photons/MAC). (a) Dark-count fraction; (b) trim error; (c) ADC resolution; (d) ablation of the training-budget distribution: the randomized band $\log \mathcal{U}[0.25, 8]$ and a fixed $\bar{n}=1$ training point both robustify, the band generalizing across the sweep.

S-VI. QUANTITATIVE COMPARISON WITH THE STATE OF THE ART

Figure S3 summarizes the positioning of this work, and Table SXI tabulates the comparison entry by entry. Literature values are as reported in the cited works; the digital-electronics row is the ~ 1 pJ/MAC ASIC reference and digital baseline accuracy used by the wireless in-physics study [4]. PILOT-Q rows are computed on the SQL-validated twin at the closed-loop operating points of Table SII; the “conventional practice” row is the same twin operated as the delocalized demonstration operates (digitally trained, statically provisioned), so the PILOT-Q gains against it are exact like-for-like. Architectures and readout classes differ across rows (free-space, single-photon-detector, delocalized fiber), so cross-row photon numbers are indicative rather than head-to-head.

S-VII. CLOSED-LOOP OPERATING POINTS AND ORACLE BOUNDS

Tables SXII–SXIV list the photon-budget Pareto points of the confidence-gated controller for each dataset, and Table SXV the corresponding oracle escalation bounds, that is, the accuracy and budget attained when exactly the mispredicted low-budget inputs are re-exposed, which no realizable per-input policy can exceed. The proximity of the realized fronts to their oracle rows, and the modest ratio between the oracle and static budgets, together establish the Letter’s central comparative claim: per-input adaptivity at the SQL is intrinsically bounded by the steepness of the accuracy–photon curve, and the margin-gated

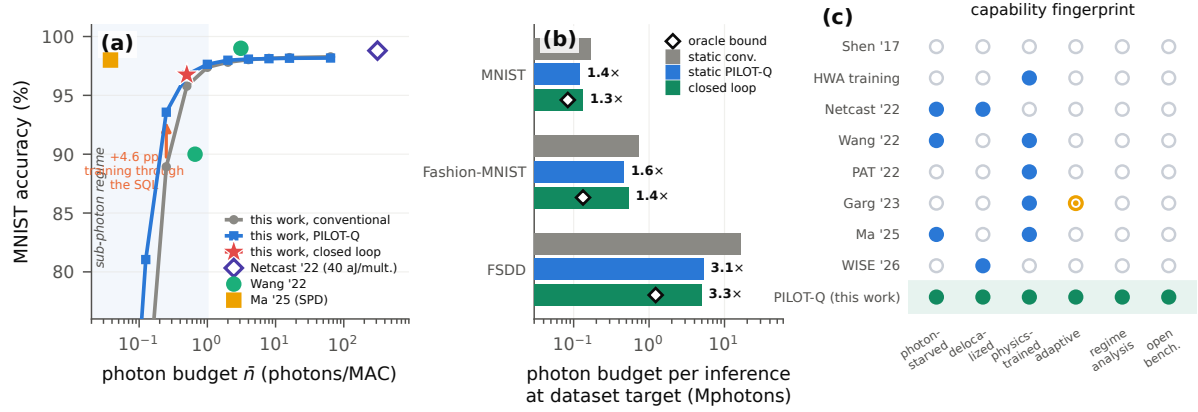


FIG. S3. Positioning against the state of the art. (a) The photon-starved landscape on MNIST: this work’s measured accuracy–photon curves under conventional and PILOT-Q training (bands: ± 1 s.d.), the closed-loop operating point at the dataset target (star), and published operating points as reported [1–3] (filled: physics-aware training; open: conventional; architectures differ, so literature points are indicative, while the curves and star are like-for-like). (b) Iso-accuracy photon budget per inference on all three benchmarks, with the oracle escalation bound (diamonds) that no per-input policy can beat. (c) Capability overview of the photon-starved and in-physics literature. Photon-starved: $\lesssim 1$ photon/MAC demonstrated; delocalized: broadcast model access; physics-trained: training through the substrate’s physics; adaptive: per-input run-time control of the physical resource; regime analysis: budget-vs-electronics accounting; open bench.: open benchmark with a validated twin. Filled = demonstrated, ringed = partial; PILOT-Q combines all six.

controller already operates near that bound.

S-VIII. TRAINING PROTOCOLS AND REPRODUCIBILITY

Table [SXVI](#) consolidates all hyperparameters. The Gaussian reparameterization carries pathwise gradients through the shot and dark noise, and the quantizer uses a straight-through estimator. All experiments run on CPU with fixed seeds, so every reported number is exactly reproducible from the released record.

TABLE SIII. Twin validation: Monte-Carlo normalized inner-product RMSE versus the analytic standard-quantum-limit prediction (400 trials at $N=4096$; 60 at $N=32,768$).

$\bar{n}$ (ph/MAC)	MC, $N=4096$	SQL, $N=4096$	MC, $N=32,768$	SQL, $N=32,768$
0.25	0.5445	0.5001	0.4886	0.5002
0.5	0.3641	0.3540	0.3433	0.3533
1	0.2461	0.2499	0.2701	0.2498
2	0.1649	0.1770	0.1903	0.1767
4	0.1342	0.1251	0.1278	0.1250
8	0.0883	0.0883	0.0888	0.0884
16	0.0639	0.0625	0.0630	0.0625
64	0.0303	0.0313	0.0303	0.0313

TABLE SIV. Photon-budget accounting per inference: received photons and optical energy, server laser energy through a 30 dB link, and the photon-independent client electronics, for the two model geometries.

$\bar{n}$	784-input model			4000-input model		
	Φ (M)	laser (pJ)	elec. (nJ)	Φ (M)	laser (pJ)	elec. (nJ)
0.25	0.067	8.5	1.23	0.308	39.4	1.23
0.5	0.133	17.1	1.23	0.616	78.9	1.23
1	0.266	34.1	1.23	1.231	157.8	1.23
2	0.532	68.2	1.23	2.462	315.5	1.23
4	1.065	136.5	1.23	4.924	631.1	1.23
8	2.130	272.9	1.23	9.848	1262.1	1.23

TABLE SV. MNIST: accuracy (% , mean $\pm$ s.d. over five noise seeds) versus photon budget on the shot-limited chain and under compound impairments ($\rho_d=20\%$, $\sigma_{cal}=15\%$, 4-bit ADC). Fixed $\bar{n}=1$: fixed-budget training ablation (PILOT-Q protocol at constant budget).

$\bar{n}$ (ph/MAC)	conventional		PILOT-Q		PILOT-Q-full		fixed $\bar{n}=1$
	shot	impaired	shot	impaired	shot	impaired	shot
0.0625	37.1 $\pm$ 0.4	26.2 $\pm$ 0.3	55.5 $\pm$ 0.6	42.5 $\pm$ 0.4	49.0 $\pm$ 0.6	39.4 $\pm$ 0.4	53.9 $\pm$ 0.7
0.125	67.4 $\pm$ 0.4	50.3 $\pm$ 0.5	81.0 $\pm$ 0.2	68.8 $\pm$ 0.6	70.2 $\pm$ 0.4	59.7 $\pm$ 0.4	82.4 $\pm$ 0.2
0.25	88.9 $\pm$ 0.2	76.4 $\pm$ 0.4	93.6 $\pm$ 0.1	87.8 $\pm$ 0.2	86.0 $\pm$ 0.3	78.6 $\pm$ 0.5	94.8 $\pm$ 0.2
0.5	95.8 $\pm$ 0.1	89.8 $\pm$ 0.3	96.8 $\pm$ 0.1	94.7 $\pm$ 0.2	92.8 $\pm$ 0.3	89.2 $\pm$ 0.3	97.3 $\pm$ 0.1
1	97.4 $\pm$ 0.1	94.2 $\pm$ 0.2	97.6 $\pm$ 0.0	96.5 $\pm$ 0.1	95.3 $\pm$ 0.1	93.4 $\pm$ 0.1	98.0 $\pm$ 0.1
2	97.8 $\pm$ 0.0	95.8 $\pm$ 0.1	98.0 $\pm$ 0.0	97.2 $\pm$ 0.0	96.2 $\pm$ 0.1	94.9 $\pm$ 0.1	98.2 $\pm$ 0.0
4	98.0 $\pm$ 0.0	96.5 $\pm$ 0.1	98.1 $\pm$ 0.0	97.5 $\pm$ 0.1	96.6 $\pm$ 0.1	95.7 $\pm$ 0.1	98.3 $\pm$ 0.0
8	98.1 $\pm$ 0.0	96.7 $\pm$ 0.2	98.1 $\pm$ 0.0	97.6 $\pm$ 0.1	96.8 $\pm$ 0.1	96.0 $\pm$ 0.1	98.3 $\pm$ 0.0
16	98.2 $\pm$ 0.0	96.9 $\pm$ 0.1	98.2 $\pm$ 0.0	97.7 $\pm$ 0.1	96.9 $\pm$ 0.0	96.1 $\pm$ 0.1	98.4 $\pm$ 0.0
64	98.3 $\pm$ 0.0	97.0 $\pm$ 0.1	98.2 $\pm$ 0.0	97.7 $\pm$ 0.1	97.0 $\pm$ 0.0	96.3 $\pm$ 0.1	98.4 $\pm$ 0.0
ideal	98.30		98.20		97.08		98.36

TABLE SVI. Fashion-MNIST: accuracy (% , mean $\pm$ s.d. over five noise seeds) versus photon budget on the shot-limited chain and under compound impairments ($\rho_d=20\%$, $\sigma_{cal}=15\%$, 4-bit ADC). Fixed $\bar{n}=1$: fixed-budget training ablation (PILOT-Q protocol at constant budget).

$\bar{n}$ (ph/MAC)	conventional		PILOT-Q		PILOT-Q-full	
	shot	impaired	shot	impaired	shot	impaired
0.0625	20.6 ± 0.3	16.1 ± 0.2	34.3 ± 0.6	26.0 ± 0.3	34.4 ± 0.5	27.9 ± 0.5
0.125	36.9 ± 0.5	27.3 ± 0.3	56.5 ± 0.2	44.4 ± 0.2	50.3 ± 0.4	41.9 ± 0.4
0.25	59.2 ± 0.3	45.2 ± 0.1	75.0 ± 0.1	64.9 ± 0.5	66.0 ± 0.3	58.5 ± 0.3
0.5	75.6 ± 0.1	62.9 ± 0.3	83.1 ± 0.3	78.0 ± 0.4	76.2 ± 0.2	71.5 ± 0.3
1	83.7 ± 0.2	73.8 ± 0.5	86.4 ± 0.2	83.6 ± 0.3	81.3 ± 0.3	78.4 ± 0.3
2	87.0 ± 0.1	79.2 ± 0.4	87.9 ± 0.1	85.9 ± 0.2	83.7 ± 0.2	81.5 ± 0.3
4	88.3 ± 0.2	81.4 ± 0.3	88.5 ± 0.1	87.1 ± 0.2	85.0 ± 0.1	82.9 ± 0.2
8	88.8 ± 0.1	82.5 ± 0.3	89.0 ± 0.1	87.6 ± 0.1	85.7 ± 0.2	83.5 ± 0.3
16	89.2 ± 0.1	83.1 ± 0.3	89.1 ± 0.1	87.8 ± 0.2	86.1 ± 0.2	83.9 ± 0.2
64	89.4 ± 0.0	83.4 ± 0.2	89.2 ± 0.1	88.0 ± 0.2	86.4 ± 0.1	84.1 ± 0.2
ideal	89.56		89.32		86.33	

TABLE SVII. FSDD: accuracy (% , mean $\pm$ s.d. over five noise seeds) versus photon budget on the shot-limited chain and under compound impairments ($\rho_d=20\%$, $\sigma_{cal}=15\%$, 4-bit ADC). Fixed $\bar{n}=1$: fixed-budget training ablation (PILOT-Q protocol at constant budget).

$\bar{n}$ (ph/MAC)	conventional		PILOT-Q		PILOT-Q-full	
	shot	impaired	shot	impaired	shot	impaired
0.0625	14.8 ± 1.0	14.3 ± 2.3	30.3 ± 1.0	26.4 ± 3.0	23.1 ± 0.6	20.4 ± 3.7
0.125	21.4 ± 1.4	19.6 ± 1.7	44.7 ± 1.6	38.4 ± 3.0	30.6 ± 1.2	28.3 ± 3.0
0.25	30.6 ± 2.0	26.2 ± 1.4	61.2 ± 2.8	54.7 ± 2.0	41.4 ± 1.8	38.9 ± 2.3
0.5	47.0 ± 1.4	39.3 ± 2.7	75.3 ± 1.7	68.7 ± 2.6	53.9 ± 1.6	49.3 ± 1.0
1	61.3 ± 1.5	51.6 ± 2.7	82.8 ± 0.7	78.6 ± 2.0	65.5 ± 1.4	59.6 ± 2.1
2	75.1 ± 1.9	61.3 ± 4.1	86.6 ± 0.4	83.3 ± 1.3	74.5 ± 0.9	67.9 ± 2.3
4	82.3 ± 1.4	69.1 ± 2.4	87.6 ± 1.1	84.9 ± 1.5	80.9 ± 1.1	72.3 ± 2.4
8	86.1 ± 1.1	72.7 ± 0.9	88.3 ± 0.9	85.7 ± 0.9	83.9 ± 1.1	75.4 ± 2.2
16	88.2 ± 0.9	73.6 ± 1.1	88.4 ± 0.9	86.3 ± 1.0	85.0 ± 0.7	76.3 ± 2.5
64	89.4 ± 0.6	74.6 ± 2.3	88.0 ± 0.5	86.4 ± 0.9	85.7 ± 0.2	76.7 ± 0.9
ideal	89.67		88.00		85.67	

TABLE SVIII. Impairment sweep at $\bar{n}=2$ photons/MAC: accuracy (%) versus dark-count fraction ρ_d .

Dataset	Training	0	0.1	0.2	0.3	0.5	1.0
MNIST	conventional	97.8	97.7	97.6	97.5	97.2	96.3
	PILOT-Q	98.0	97.9	97.9	97.8	97.6	97.2
	PILOT-Q-full	96.2	96.0	95.9	95.7	95.3	94.2
Fashion-MNIST	conventional	87.0	86.2	85.4	84.6	82.8	78.1
	PILOT-Q	87.9	87.5	87.2	86.8	86.1	84.2
	PILOT-Q-full	83.7	83.1	82.5	82.0	80.7	77.8
FSDD	conventional	75.1	71.3	68.3	65.6	60.5	52.4
	PILOT-Q	86.6	85.8	85.1	84.6	81.9	77.9
	PILOT-Q-full	74.5	72.5	70.0	68.5	65.4	58.3

TABLE SIX. Impairment sweep at $\bar{n}=2$ photons/MAC: accuracy (%) versus trim error σ_{cal} .

Dataset	Training	0	0.05	0.10	0.15	0.20	0.30
MNIST	conventional	97.8	97.8	97.8	97.7	97.5	97.1
	PILOT-Q	98.0	97.9	97.9	97.8	97.8	97.6
	PILOT-Q-full	96.2	96.3	96.2	96.2	96.1	95.9
Fashion-MNIST	conventional	87.0	86.7	86.5	86.1	85.5	83.6
	PILOT-Q	87.9	88.0	87.9	87.8	87.6	87.1
	PILOT-Q-full	83.7	83.9	83.7	83.6	83.3	82.8
FSDD	conventional	75.1	73.9	73.5	72.6	69.9	63.5
	PILOT-Q	86.6	86.1	86.5	86.5	86.7	85.6
	PILOT-Q-full	74.5	76.2	76.2	75.3	74.3	72.9

TABLE SX. Impairment sweep at $\bar{n}=2$ photons/MAC: accuracy (%) versus ADC resolution (bits).

Dataset	Training	2	3	4	5	6	8
MNIST	conventional	59.3	93.3	96.6	97.6	97.8	97.8
	PILOT-Q	89.1	96.3	97.5	97.8	97.9	98.0
	PILOT-Q-full	86.9	93.8	95.6	96.0	96.2	96.2
Fashion-MNIST	conventional	28.3	62.0	81.2	85.8	86.7	87.0
	PILOT-Q	57.9	81.8	86.8	87.6	87.8	87.9
	PILOT-Q-full	48.5	74.7	82.4	83.6	83.9	83.9
FSDD	conventional	17.8	54.2	67.9	72.3	74.3	75.0
	PILOT-Q	63.7	78.5	83.2	86.2	86.0	86.1
	PILOT-Q-full	45.6	67.9	70.6	73.7	75.0	74.9

TABLE SXI. Quantitative figure-of-merit comparison for photon-starved and in-physics inference. Bracketed numbers refer to the reference list of the main document.

Scheme	Architecture	Task	Acc. (%)	Photons/MAC (or energy)	Key figure of merit
Digital electronics [4]	ASIC reference	MNIST	98.1	~ 1 pJ/MAC	baseline used by WISE
Netcast [1] (2022)	delocalized fiber, integrating client	MNIST	98.8	40 aJ/mult. (< 1 ph w/ SPDs)	86 km deployment, 98.8%
Wang <i>et al.</i> [2] (2022)	free-space, noise-aware training	MNIST	99 / 90	3.1 / 0.66	sub-photon multiplication
Ma <i>et al.</i> [3] (2025)	SPD, stochastic binary act.	MNIST	98	0.038	SNR ≈ 1 operation
WISE [4] (2026)	wireless RF mixer	MNIST	95.7	6.0 fJ/MAC	disaggregated RF MVMs
Conventional practice (twin)	delocalized twin (this work)	MNIST	96.3	0.63	digitally trained, static budget
PILOT-Q closed loop	delocalized twin (this work)	MNIST	96.8	0.50	$1.3\times$ fewer photons @ iso-acc.
PILOT-Q closed loop		Fashion-MNIST	87.9	2.00	$1.4\times$ fewer photons @ iso-acc.
PILOT-Q closed loop		FSDD	87.7	4.04	$3.3\times$ fewer photons @ iso-acc.

TABLE SXII. Photon-budget Pareto points of the closed-loop controller on MNIST (PILOT-Q model), thinned to ≥ 0.25 percentage-point accuracy increments (complete sweeps in the released record). Φ : photons per inference.

$\bar{n}_{lo}$	$\bar{n}_{hi}$	τ	acc. (%)	esc. (%)	Φ (Mphotons)
0.25	1	0	93.57	0.0	0.067
0.5	2	0	96.75	0.0	0.133
1	4	0	97.64	0.0	0.266
2	8	0	97.99	0.0	0.532

TABLE SXIII. Photon-budget Pareto points of the closed-loop controller on Fashion-MNIST (PILOT-Q model), thinned to ≥ 0.25 percentage-point accuracy increments (complete sweeps in the released record). Φ : photons per inference.

$\bar{n}_{lo}$	$\bar{n}_{hi}$	τ	acc. (%)	esc. (%)	Φ (Mphotons)
0.25	1	0	74.96	0.0	0.067
0.25	1	0.5	75.23	0.7	0.068
0.25	1	0.8	75.53	1.5	0.070
0.5	2	0	83.12	0.0	0.133
0.5	4	0.8	83.38	1.1	0.145
1	4	0	86.39	0.0	0.266
2	8	0	87.87	0.0	0.532
4	16	0	88.55	0.0	1.065

TABLE SXIV. Photon-budget Pareto points of the closed-loop controller on FSDD (PILOT-Q model), thinned to ≥ 0.25 percentage-point accuracy increments (complete sweeps in the released record). Φ : photons per inference.

$\bar{n}_{lo}$	$\bar{n}_{hi}$	τ	acc. (%)	esc. (%)	Φ (Mphotons)
0.25	1	0	61.20	0.0	0.308
0.25	1	0.2	61.53	0.9	0.318
0.25	1	0.3	62.07	1.5	0.327
0.25	1	0.5	62.67	2.6	0.340
0.25	1	0.6	63.13	3.8	0.355
0.25	1	0.8	63.93	6.1	0.383
0.5	2	0	75.33	0.0	0.616
0.5	2	0.2	75.67	1.0	0.640
0.5	2	0.4	76.00	1.6	0.655
0.5	2	0.6	76.60	2.6	0.680
0.5	2	0.8	76.87	4.4	0.724
1	4	0	82.80	0.0	1.231
1	4	0.2	83.07	0.7	1.267
1	4	0.4	83.67	2.0	1.329
1	4	0.8	84.07	4.3	1.444
2	8	0	86.60	0.0	2.462
2	8	0.8	87.00	3.2	2.777
4	16	0	87.60	0.0	4.924

TABLE SXV. Oracle escalation bounds (PILOT-Q models): accuracy and photon budget when exactly the inputs mispredicted at $\bar{n}_{lo}$ are re-exposed at $\bar{n}_{hi}$, the upper envelope of any per-input policy.

Dataset	$\bar{n}_{lo}$	$\bar{n}_{hi}$	acc. (%)	esc. (%)	Φ (Mphotons)
MNIST	0.25	1	98.24	6.4	0.084
	0.5	2	98.40	3.2	0.150
	0.5	4	98.50	3.2	0.168
	1	4	98.42	2.4	0.291
	2	8	98.38	2.0	0.575
	4	16	98.34	1.9	1.147
Fashion-MNIST	0.25	1	89.34	25.0	0.133
	0.5	2	90.50	16.9	0.223
	0.5	4	91.07	16.9	0.313
	1	4	90.69	13.6	0.411
	2	8	90.54	12.1	0.791
	4	16	90.29	11.5	1.553
FSDD	0.25	1	86.87	38.8	0.785
	0.5	2	90.00	24.7	1.223
	0.5	4	91.13	24.7	1.830
	1	4	90.87	17.2	2.078
	2	8	90.93	13.4	3.782
	4	16	90.33	12.4	7.366

TABLE SXVI. Training protocols and hyperparameters. All protocols share: Adam, learning rate 10^{-3} decayed $\times 0.3$ at 70% of training, batch 256, training seed 42, evaluation seeds $\{101, 202, 303, 404, 505\}$.

	conventional	PILOT-Q	PILOT-Q-full	PILOT-Q (fixed)
forward physics	clean	twin	twin	twin
$\bar{n}$ (ph/MAC) ^a	—	$\log \mathcal{U}[0.25, 8]$	$\log \mathcal{U}[0.25, 8]$	1
dark fraction ρ_d	—	0	$\mathcal{U}[0, 0.20]$	0
trim error σ_{cal}	—	0	$\mathcal{U}[0, 0.15]$	0
ADC bits b	—	∞	$\{4, 6, 8, \infty\}$	∞
clean warmup (epochs)	—	15%	15%	15%
clean batch anchor	—	35%	35%	35%
epochs (image / audio)	25 / 100	35 / 140	35 / 140	35 / —

^a $\log \mathcal{U}[0.5, 8]$ on Fashion-MNIST, whose harder decision boundary tolerates less training noise (see the training-protocol description in the main document).